MATH 1271 **Name:** _____
Fall 2026
Quiz 9: 10/13 Content
10/15/2026
Time Limit: 5 minutes **Student ID:** _____

This quiz consists of 1 questions across 2 pages worth 5 points.
Rules:

- You have the indicated amount of time above to complete the quiz and you must not begin working on the quiz until instructed to do so.
- You must show all of your work to receive full credit, unless stated otherwise. **On problems that require work to be shown, answers that are mysterious or unsupported will not receive full credit.**
- You may not use a calculator, textbook, notes or any other external resources.
- Please write your initials at the top of every page in the event the pages become separated.
- If you run out of space on a page, feel free to use the back of the page to continue your work.
- You understand that if you violate any of these or the University's rules, you will receive a score of zero on this quiz and you may be subject to further disciplinary action.

Following the University of Minnesota's policy on academic honesty, I pledge that the work shown on this quiz is entirely my own and that I have followed the rules stated above.

Signature: _____

Question	Points
1	5
Total:	5

1. (5 points) Compute the derivative of

$$p(q) = \frac{q^3}{\sin(q) + 6q^8}. \tag{1}$$

You need not expand the denominator of the resulting function.

Solution:

$$\frac{d}{dq}p(q) = \frac{d}{dq}\frac{q^3}{\sin(q) + 6q^8} \tag{2}$$

$$= \frac{\frac{d}{dq}(q^3) \cdot (\sin(q) + 6q^8) - q^3 \cdot \frac{d}{dq}(\sin(q) + 6q^8)}{(\sin(q) + 6q^8)^2} \tag{3}$$

$$= \frac{3q^2 \cdot (\sin(q) + 6q^8) - q^3 \cdot (\cos(q) + 48q^7)}{(\sin(q) + 6q^8)^2} \tag{4}$$

$$= \frac{3q^2 \sin(q) + 18q^{10} - q^3 \cos(q) + 48q^{10}}{(\sin(q) + 6q^8)^2} \qquad = \frac{3q^2 \sin(q) - q^3 \cos(q) + 64q^{10}}{(\sin(q) + 6q^8)^2} \tag{5}$$